POWDER METALLURGY

Complete Study Notes

Metal powder production, powder characteristics, blending, compaction, pre-sintering, sintering, secondary operations, applications, advantages and limitations

Reference framework: S. K. Hajra Choudhury, A. K. Hajra Choudhury and Nirjhar Roy, Elements of Workshop Technology, Volume I - Manufacturing Processes. These notes are paraphrased and organized for study and examination preparation.

Topics Covered

- 1. Introduction to Powder Metallurgy
- 2. Methods of producing metal powders
- 3. Characteristics of metal powders
- 4. Blending and mixing
- 5. Briquetting / compacting
- 6. Pre-sintering
- 7. Sintering
- 8. Secondary operations: sizing, coining, machining and related treatments
- 9. Applications of Powder Metallurgy
- 10. Advantages of Powder Metallurgy
- 11. Disadvantages and limitations of Powder Metallurgy
- 12. Quick revision tables and viva questions

1. Introduction to Powder Metallurgy

Definition. Powder metallurgy (P/M) is a manufacturing process in which metallic powders are produced, conditioned and blended, compacted into a desired shape under pressure, and then heated to a temperature below the melting point of the principal constituent so that the particles bond together. The heating operation is called sintering.

The process is especially valuable when parts are required in large numbers, when materials are difficult to shape by conventional melting and machining, or when controlled porosity, special compositions, or near-net-shape production is desired.

Basic sequence of the P/M process

1. Production of suitable metal powder.
2. Sieving, classification and conditioning of the powder.
3. Mixing or blending with alloying powders, lubricants or other additions.
4. Compacting the powder in a die to obtain a green compact or briquette.
5. Pre-sintering when required for handling or machining before final sintering.
6. Sintering in a controlled atmosphere at a suitable temperature and time.
7. Carrying out secondary operations such as sizing, coining, machining, impregnation, infiltration, heat treatment, plating or grinding where required.

Key idea: Powder metallurgy generally avoids complete melting of the main metal. Shape is first produced by pressure, and strength is developed mainly by sintering.

2. Methods of Producing Metal Powders

The method of powder production strongly influences particle shape, particle size, purity, apparent density, compressibility and the final properties of the compact. No single powder-making process is suitable for every metal.

Method	Principle	Powder characteristics	Typical use
Mechanical pulverization / comminution	Bulk metal is crushed, ground, milled or otherwise mechanically broken into smaller particles. Brittle metals and alloys can be powdered relatively easily by crushing or ball milling. Ductile metals tend to flatten rather than fracture.	Irregular or flaky particles; often work-hardened; contamination from mill media is possible.	Brittle metals/alloys, some ferroalloys, powders requiring flaky shape.
Atomization	A stream of molten metal is broken into fine droplets by a high-velocity jet of air, inert gas or water. The droplets rapidly solidify into powder.	Gas atomization tends to produce smoother, more spherical particles; water atomization generally gives more irregular particles.	Iron, steel, aluminium, copper, brass and many alloy powders.
Reduction of metal oxides	A metal oxide is heated in a reducing atmosphere such as hydrogen or carbon monoxide. Oxygen is removed and a porous or sponge-like metallic mass is obtained; it is then crushed and classified.	Porous, irregular particles with good compacting characteristics.	Iron, copper, tungsten, molybdenum and related powders.
Electrolytic deposition	Metal is deposited electrolytically on the cathode from a suitable electrolyte. The deposit is stripped, washed, dried, annealed if necessary, and pulverized.	High-purity, dendritic or spongy powder; good compressibility but relatively costly.	Copper, iron, nickel and other high-purity powders.
Carbonyl process	Certain metals form volatile carbonyl compounds with carbon monoxide. Controlled decomposition of the carbonyl yields very fine metallic powder.	Very fine, high-purity, nearly spherical powder.	Iron and nickel powders.
Chemical precipitation / decomposition	A dissolved metal compound is chemically reduced or precipitated, or a compound is thermally decomposed to obtain fine metal powder.	Very fine powders; purity can be high if process control is good.	Silver, copper and other special powders.
Condensation from vapour	Metal is vaporized and then condensed under controlled conditions.	Extremely fine, often spherical or near-spherical particles; expensive.	Special fine powders for electronics, pigments and research.
Shotting / granulation	Molten metal is poured through an orifice, sieve or perforated plate and broken into droplets that solidify while falling through a cooling medium.	Relatively coarse, rounded particles.	Low-melting metals and coarse metallic granules.

2.1 Mechanical methods

In mechanical production, external forces are used to fracture the material. Crushing, grinding, hammer milling and ball milling are common approaches. Mechanical methods are most successful for brittle materials because they break readily. Very ductile metals usually deform into flakes instead of forming equiaxed particles.

- Simple equipment and suitable for brittle materials.
- Particle shape is usually irregular, which can improve interlocking during compaction.
- Possible disadvantages are contamination, wide particle-size distribution, work hardening and high power consumption for very fine powders.

2.2 Atomization

Atomization is one of the most important commercial methods. The molten metal is forced through a small opening and disintegrated by a high-speed fluid jet. Each small droplet cools and solidifies before collection.

- Water atomization: rapid cooling; generally irregular particles; economical for many ferrous powders.
- Gas atomization: smoother and more spherical particles; useful where flowability and cleanliness are important.
- Centrifugal atomization: molten metal is thrown outward from a rotating surface or electrode to form droplets.

2.3 Reduction method

In this method, metal oxides are reduced at elevated temperature. The resulting sponge metal is crushed and screened. The porous structure provides good compressibility and gives a powder that bonds well during sintering.

2.4 Electrolytic method

Electrolysis yields powders of high chemical purity. A brittle or dendritic cathode deposit is obtained and subsequently processed into powder. Because electrical energy and chemical handling are required, the process is generally more expensive than bulk mechanical or atomization methods.

3. Characteristics of Metal Powders

The behaviour of a metal powder during mixing, die filling, compacting and sintering depends strongly on its physical and chemical characteristics. These characteristics also influence density, dimensional accuracy, strength and surface quality of the final part.

Characteristic	Meaning and importance
Particle size	Average dimension of the individual particles. Fine particles generally sinter more readily because of larger surface area, but may have poor flow and higher oxidation tendency.
Particle-size distribution	Relative amount of coarse and fine particles in the powder. A well-selected distribution can improve packing density because smaller particles fill voids between larger particles.
Particle shape	May be spherical, rounded, irregular, angular, flaky, dendritic or porous. Shape affects flowability, packing, interlocking and compressibility.
Specific surface / surface area	Surface area per unit mass. Finer and more irregular particles have larger surface area and therefore higher surface energy and sintering activity.
Apparent density	Mass of loose powder divided by the volume it occupies without deliberate compaction. It indicates how densely the powder naturally packs.
Tap density	Density after a specified vibration or tapping treatment. It indicates the improved packing possible when particles rearrange.
Flow rate / flowability	Ability of powder to flow freely through an orifice and fill a die uniformly. Spherical particles generally flow better than irregular fine particles.
Compressibility	Ability of the powder to decrease in volume and achieve high density under pressure.
Compactability	Ability to form a coherent green compact of sufficient strength at a given pressure.
Green strength	Mechanical strength of the compact before sintering. It must be sufficient for ejection, handling and transfer to the furnace.
Purity / chemical composition	Impurities and oxide films can interfere with sintering and alter properties. Chemical composition must be controlled for alloy powders.
Porosity	Voids within or between powder particles. Porosity affects density, compressibility, sintering and final permeability.

Sinterability	Ability of particles to form strong bonds and densify during sintering at a chosen temperature and atmosphere.

3.1 Effect of particle size and shape

- Fine powder gives a large contact area and generally promotes rapid sintering, but oxidation and poor flow can become serious.
- Coarse powder flows well and is easier to handle but may give larger pores and require stronger sintering conditions.
- Spherical powders usually show excellent flow and high apparent density.
- Irregular and dendritic powders interlock strongly and often provide good green strength.
- A broad, controlled particle-size distribution can improve packing by filling the spaces between coarse particles with finer ones.

4. Blending and Mixing

Blending is the operation of thoroughly intermixing powders so that the final charge has uniform composition and consistent processing behaviour. The terms mixing and blending are often used together; in practical P/M work, mixing may include powders of different materials as well as lubricants, graphite or other additions.

Objectives of blending

- To obtain a uniform chemical composition throughout the batch.
- To mix powders of different particle sizes so that packing characteristics are improved.
- To distribute lubricants uniformly and reduce die-wall friction during compacting.
- To add alloying elements such as graphite, copper or nickel in controlled amounts.
- To improve flow, compressibility and consistency from one compact to another.

Common blending equipment

- Double-cone blender
- V-blender
- Drum or barrel mixer
- Ribbon blender
- Ball mill for certain powder systems when simultaneous size reduction is acceptable

Blending time should be sufficient for uniformity but should not be excessive. Overmixing can cause particle segregation, excessive cold welding in ductile powders, lubricant smearing or unwanted size reduction.

Exam point: Blending must produce a homogeneous mixture without damaging particle shape or causing segregation.

5. Briquetting / Compacting

Briquetting, also called compacting or pressing, is the operation of compressing the blended powder in a die under pressure to form a coherent shape called a green compact or briquette. The compact has the approximate geometry of the final part but has not yet developed full strength.

Purposes of compacting

- To give the powder the required shape and dimensions.
- To increase particle-to-particle contact area.
- To produce sufficient green strength for safe ejection and handling.

- To reduce porosity and raise density to a desired level.
- To prepare the compact for efficient bonding during sintering.

Stages during die compaction

8. Powder is filled into the die cavity.
9. Pressure is applied by one or more punches.
10. Particles rearrange and fill large voids.
11. Further pressure causes elastic and plastic deformation and some particle fracture, increasing contact area.
12. The pressure is released and the green compact undergoes slight elastic spring-back.
13. The compact is ejected carefully from the die.

Methods of compaction

Method	Description / use
Single-action pressing	Pressure is applied mainly from one punch. Density may be less uniform in thick parts because die-wall friction reduces pressure transmission.
Double-action pressing	Top and bottom punches move so that pressure is applied from both directions, giving better density uniformity.
Isostatic pressing	Powder is enclosed in a flexible mould and subjected to nearly uniform fluid pressure from all directions. Useful for larger or complex shapes.
Hot pressing	Compaction and heating are applied simultaneously to improve densification. Used for difficult materials and high-performance products.
Roll compaction	Powder is fed between rolls to produce strip or sheet-like compacts.
Extrusion of powder	Powder or a powder-filled container is forced through a die, sometimes after preliminary compaction.

Factors affecting compact density

- Compacting pressure
- Particle size and distribution
- Particle shape
- Powder hardness and ductility
- Lubricant content
- Die-wall friction
- Geometry and thickness of the part
- Method of pressing and number of punches

A major practical issue is non-uniform density. The region close to the moving punch tends to experience higher pressure than regions farther away. Double-action pressing and suitable die design are therefore used to improve density distribution.

6. Pre-sintering

Pre-sintering is a preliminary heating treatment given to a green compact at a temperature lower than the final sintering temperature. Its purpose is to develop enough strength for further handling or intermediate operations without completing the full sintering process.

Why pre-sintering is used

- To increase strength so that a fragile green compact can be handled safely.
- To permit machining of features that are difficult or impossible to form directly in the die.
- To burn off lubricants and volatile additions before final sintering.
- To reduce the risk of cracking or distortion in a subsequent treatment.

- To prepare multi-stage or complicated compacts for final densification.

After pre-sintering, operations such as drilling, turning or shaping may be carried out more safely than on an unsintered compact. The component is then subjected to final sintering.

7. Sintering

Sintering is the controlled heating of a green compact to a temperature below the melting point of the principal constituent, usually in a protective or controlled atmosphere, for sufficient time to develop metallurgical bonding between powder particles. It is the most important heat-treatment stage of the conventional powder-metallurgy route.

Purpose of sintering

- To create strong metallurgical bonds between adjacent powder particles.
- To increase strength, hardness and toughness compared with the green compact.
- To reduce internal stresses introduced during compaction.
- To improve electrical and thermal conductivity.
- To permit diffusion and alloy homogenization.
- To change pore shape and, depending on conditions, reduce porosity and increase density.

Stages of sintering

14. Initial heating: moisture and volatile substances are removed; lubricants may burn off or vaporize under controlled conditions.
15. Bond formation: necks begin to form at points of contact between adjacent particles because atoms diffuse across the interfaces.
16. Neck growth and pore rounding: particle bonds become wider and stronger; pores become more rounded and may shrink.
17. Diffusion and homogenization: alloying elements diffuse, composition becomes more uniform, and the compact develops final mechanical properties.
18. Controlled cooling: the part is cooled under conditions that avoid oxidation, distortion or unwanted microstructural changes.

Sintering temperature and time

The sintering temperature is selected as a substantial fraction of the absolute melting temperature of the principal metal. It must be high enough for rapid atomic diffusion but below the temperature at which the whole compact melts. The exact temperature and time depend on the material, powder size, density, alloying additions and required properties.

Sintering atmosphere

Atmosphere	Function / reason for use
Hydrogen	Strongly reducing; helps remove oxides and is used for certain steels, copper, tungsten and other powders.
Dissociated ammonia	Mixture mainly of hydrogen and nitrogen; provides a reducing/protective atmosphere for many ferrous parts.
Endothermic / exothermic gas	Prepared furnace atmospheres used industrially to control oxidation and carbon potential.
Nitrogen / inert gas	Provides protection against oxidation for suitable alloys; argon is used where high inertness is required.
Vacuum	Used for reactive, high-purity or special materials where gaseous contamination must be minimized.

Key concept: Sintering bonds particles primarily by diffusion and related mass-transport mechanisms. The compact becomes much stronger without completely melting the main metal.

8. Secondary Operations

Although many powder-metallurgy parts leave the sintering furnace close to final size, secondary operations may be required to improve dimensional accuracy, density, surface finish, mechanical properties or functional performance.

Operation	Purpose	Typical use
Sizing	The sintered part is repressed in a precision die mainly to correct dimensions and improve tolerances. Only a small amount of deformation is involved.	Bushes, gears, structural components requiring close size control.
Coining	A heavier repressing operation than sizing. Considerable pressure is applied to improve density, surface finish, dimensional accuracy and sometimes strength.	Dense surfaces, accurate profiles, gear teeth and components requiring improved strength.
Machining	Turning, drilling, tapping, milling, grinding or other cutting operations are used when features cannot be pressed directly or when exceptionally close tolerances are needed.	Threads, cross holes, undercuts, grooves and precision faces.
Impregnation	Open pores are filled with a non-metallic substance such as lubricating oil, resin or wax.	Self-lubricating bearings and sealing of porous parts.
Infiltration	A lower-melting metal is melted into the interconnected pores of a sintered compact by capillary action.	Increasing density, strength, toughness or thermal/electrical conductivity.
Heat treatment	Carburizing, quenching, tempering, steam treatment or other heat treatments may be used to improve hardness, wear resistance or corrosion behaviour.	P/M gears, cams and structural steel parts.
Surface finishing / plating	Grinding, lapping, polishing, plating or coating improves surface finish, corrosion resistance or appearance.	Precision and decorative parts.
Repressing	A sintered part is subjected to another pressing step to raise density or improve shape; it may be followed by resintering.	High-strength structural components.

8.1 Sizing

Sizing is primarily a dimensional correction operation. The sintered component is forced into a sizing die that is very close to the desired final dimensions. The operation improves tolerances and restores roundness or alignment with little intentional change in density.

8.2 Coining

Coining uses higher pressure than sizing and deliberately plastically deforms the sintered part. It may increase density locally or throughout the component, sharpen details, improve surface finish and enhance dimensional accuracy.

8.3 Machining

Powder-metallurgy processing is selected partly to reduce machining, but machining cannot always be eliminated. Features transverse to the pressing direction, threads, deep narrow holes, undercuts and very close-tolerance surfaces are commonly machined after sintering or after pre-sintering.

Difference to remember: Sizing mainly corrects dimensions; coining applies heavier pressure and can improve density and surface detail as well as dimensions.

9. Applications of Powder Metallurgy

Powder metallurgy is used where its ability to form near-net shapes, control porosity, combine otherwise difficult materials, or economically manufacture large quantities gives an advantage over casting or machining.

Application	Why P/M is suitable
Self-lubricating bearings and bushes	Porosity is intentionally retained and filled with lubricating oil. During operation, oil reaches the bearing surface by capillary action.
Gears, sprockets, cams and structural parts	High production rate and near-net-shape manufacture make P/M economical for repeated small and medium-sized components.
Electrical contacts and brushes	Materials such as copper-graphite, silver-based contact materials and other composites can be made by powder routes.
Cemented carbides	Hard tungsten carbide particles are bonded with cobalt or another metallic binder to produce cutting-tool and wear-resistant materials.
Porous filters	Controlled interconnected porosity allows liquids or gases to pass while solid particles are trapped.
Refractory-metal products	Tungsten, molybdenum and related high-melting materials are difficult to cast conventionally and are commonly processed from powder.
Permanent magnets and magnetic components	Ferrites and special magnetic alloys can be compacted into useful shapes with controlled composition.
Friction materials	Brake and clutch materials can combine metallic powders with graphite, ceramics and other constituents.
Nuclear and high-temperature materials	Special powders allow controlled compositions and fabrication of materials that may be difficult to melt or cast.
Diamond-metal tools	Diamond particles can be bonded in a metallic matrix for grinding, cutting and drilling tools.
Automotive components	Valve-seat inserts, connecting-rod parts, synchronizer hubs, oil-pump gears and many other high-volume parts are produced by P/M.
Additive manufacturing feedstock	Modern metal additive manufacturing also uses carefully controlled metal powders, especially spherical atomized powders, although the consolidation route differs from conventional die-pressed P/M.

10. Advantages of Powder Metallurgy

- Very high material utilization because little or no machining scrap is produced.
- Near-net-shape production reduces subsequent machining and finishing.
- Excellent suitability for mass production of small and medium components with repeatable dimensions.
- Ability to manufacture materials with very high melting temperatures, such as tungsten and molybdenum.
- Ability to combine materials that may be difficult to alloy by conventional melting, for example metal-graphite or carbide-metal composites.
- Controlled porosity can be deliberately produced for self-lubricating bearings, filters and flow-control components.
- Good control of chemical composition and possibility of producing special alloys and composites.
- Good dimensional consistency when tooling and powder conditions are controlled.
- Surface finish can be good directly after pressing and sintering.
- Automation of powder filling, pressing and furnace processing is relatively straightforward for large production runs.
- Energy may be saved for certain products because large amounts of metal do not need to be fully melted and repeatedly machined.
- Secondary heat treatment, infiltration, impregnation and coining provide flexibility in tailoring properties.

11. Disadvantages and Limitations of Powder Metallurgy

- Powder production can be expensive, especially for very fine, very pure or spherical powders.
- Presses, precision dies and tooling involve high initial cost; therefore the process is less economical for very small production quantities.
- Component size and shape are limited by the capacity of the press and by the need to eject the compact from the die.
- Features such as severe undercuts, re-entrant shapes, transverse holes and very complex side details are difficult to form directly by conventional die pressing.
- Density may be non-uniform because pressure is lost through die-wall friction, particularly in thick or complicated parts.
- Residual porosity usually means mechanical properties such as ductility, impact strength and fatigue strength may be lower than those of fully dense wrought material unless additional densification is used.
- Powders can oxidize, absorb moisture or become contaminated during storage and handling.
- Fine metal powders can create dust, fire or explosion hazards and require careful housekeeping and safety controls.
- Dimensional changes can occur during sintering because of shrinkage or growth; careful process control is needed.
- Material selection may be limited by availability and cost of suitable powder.
- Large parts may require very high compacting forces and expensive equipment.
- Joining, machining or finishing may still be required when the part contains shapes that cannot be pressed directly.

11.1 Practical limitations in component design

- Uniform wall thickness and a clear pressing direction are desirable.
- Sharp internal corners should be avoided where possible because they make die filling and tooling more difficult.
- Long thin sections and abrupt section changes can give non-uniform density or cracking.
- Holes parallel to the pressing direction are comparatively easy; cross holes generally require secondary machining.
- The part must normally be capable of being ejected from a rigid die without damage.

12.

Stage	Main purpose
Powder production	Obtain metal in the required particle size, shape and purity.
Blending	Achieve uniform composition and add lubricants/alloying additions.
Briquetting / compaction	Form the green shape and create sufficient density and green strength.
Pre-sintering	Give preliminary strength for handling or intermediate machining.
Sintering	Develop metallurgical bonding and final structural strength.
Sizing	Correct dimensions and improve tolerances.
Coining	Apply heavier repressing to improve density, finish and accuracy.
Machining	Produce features or tolerances not obtainable directly by pressing.
Impregnation	Fill pores with a non-metal such as oil.
Infiltration	Fill pores with a molten metal to raise density/properties.

Term	One-line recall
Apparent density	Mass of loose powder divided by its bulk volume.
Flowability	Ability of powder to flow and fill a die uniformly.

Compressibility	Ability of powder to increase density under pressure.
Green compact	Pressed powder shape before sintering.
Green strength	Strength of the unsintered compact.
Sintering	Heating below the main melting point to bond powder particles.
Sizing	Light repressing for dimensional correction.
Coining	Heavier repressing for density, detail and dimensional improvement.
Impregnation	Filling pores with oil/resin or another non-metal.
Infiltration	Filling pores with a lower-melting metal.

13. Short Viva / Examination Questions

19. Define powder metallurgy and list its basic process steps.
20. Why is powder metallurgy called a near-net-shape process?
21. Explain any four methods of producing metal powders.
22. Compare atomization, oxide reduction and electrolytic powder production.
23. List the important characteristics of metal powders.
24. How do particle size and particle shape affect flowability and compaction?
25. Define apparent density, compressibility and green strength.
26. What is the purpose of blending? Why are lubricants added?
27. Define briquetting or compacting. Explain the stages of die compaction.
28. Why can density vary within a pressed compact?
29. What is pre-sintering and when is it used?
30. Define sintering. Why is a controlled atmosphere required?
31. What changes take place in a compact during sintering?
32. Differentiate between sizing and coining.
33. Why may machining still be required for a P/M component?
34. Differentiate between impregnation and infiltration.
35. List important applications of powder metallurgy.
36. Why is P/M used for self-lubricating bearings?
37. State the advantages of powder metallurgy.
38. State the disadvantages and design limitations of powder metallurgy.

Reference

Primary reference framework: Hajra Choudhury, S. K.; Hajra Choudhury, A. K.; and Nirjhar Roy, S. K., Elements of Workshop Technology, Volume I - Manufacturing Processes, Media Promoters & Publishers. The wording in these notes is original and condensed for learning; the topic organization and terminology follow standard workshop/manufacturing technology treatment consistent with this reference.

Supplementary verification of the syllabus coverage was made against university course outlines that list powder production methods, powder characteristics, mixing/blending, briquetting, sintering, pre-sintering, secondary processes, applications, and advantages/disadvantages alongside Hajra Choudhury as a recommended reference.